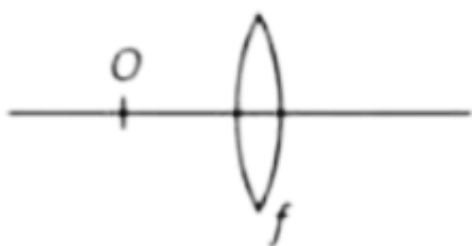


JEE-Main-23-01-2026 (Memory Based)
[EVENING SHIFT]
Physics

Question: When an object is kept at distance 12 cm and 24 cm from a convex lens, magnitude of magnification is same in both cases. Find focal length of the lens.



Answer: (6 cm)

Solution:

Ans Given $u_1 = -12\text{ cm}$ & $u_2 = -24\text{ cm}$
 and, $m_1 = m_2$, $f = ?$
 As we know, $m = \frac{v}{u}$
 $m_1 = m_2$
 $\frac{v_1}{12} = \frac{v_2}{24}$
 $\boxed{v_1 = \frac{v_2}{2}}$
 $\frac{1}{f} = \frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{v_2} - \frac{1}{u_2}$
 $\frac{1}{v_1} + \frac{1}{12} = \frac{1}{2v_1} + \frac{1}{24}$
 $-\frac{1}{v_1} = -\frac{1}{12} \Rightarrow \boxed{v_1 = 12}$
 $\therefore \frac{1}{f} = \frac{1}{12} + \frac{1}{12} = \frac{2}{12}$
 $\boxed{f = 6\text{ cm}}$

Question: A skydiver jumps from a hovering plane from a height H above the ground. He falls freely under gravity for 2 s. Then he opens his parachute and his downward motion is uniformly retarded with magnitude 3 m s^{-2} until he reaches the ground with a speed of 5 m s^{-2} . The height H is: ($g = 10\text{ m/s}^2$)

Options:

(a) 75 m

- (b) 80 m
(c) 82.5 m
(d) 90 m

Answer: (c)

Solution:

Concise Solution

Free fall for 2 s (starting from rest):

$$v_1 = gt = 10 \times 2 = 20 \text{ m s}^{-1}$$

$$s_1 = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 2^2 = 20 \text{ m}$$

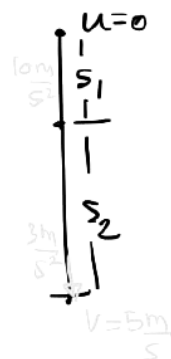
Parachute phase: take downward as positive. Retardation $3 \Rightarrow a = -3 \text{ m s}^{-2}$, landing speed $v_2 = 5 \text{ m s}^{-1}$.

$$v_2^2 = v_1^2 + 2as_2 \Rightarrow 25 = 400 + 2(-3)s_2 \Rightarrow 25 = 400 - 6s_2$$

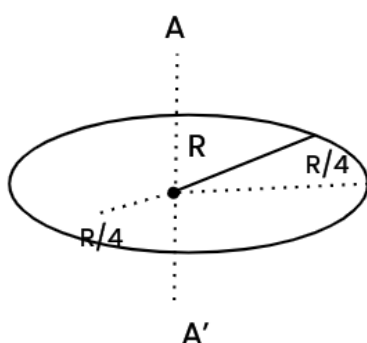
$$s_2 = \frac{375}{6} = 62.5 \text{ m}$$

Total height:

$$H = s_1 + s_2 = 20 + 62.5 = 82.5 \text{ m}$$

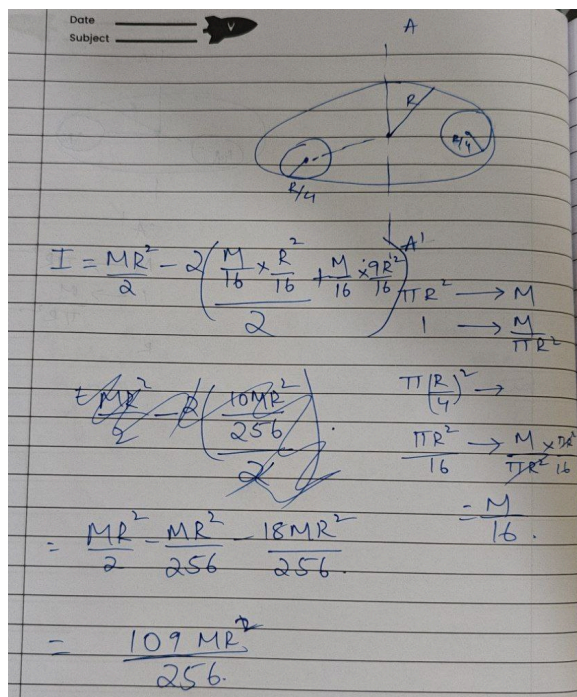


Question: From a uniform disc of radius R and mass M two small discs of radius $R/4$ is being cut as shown in figure. Find the moment of inertia of the system about axis AA' .



Answer: $\left(\frac{109MR^2}{256} \right)$

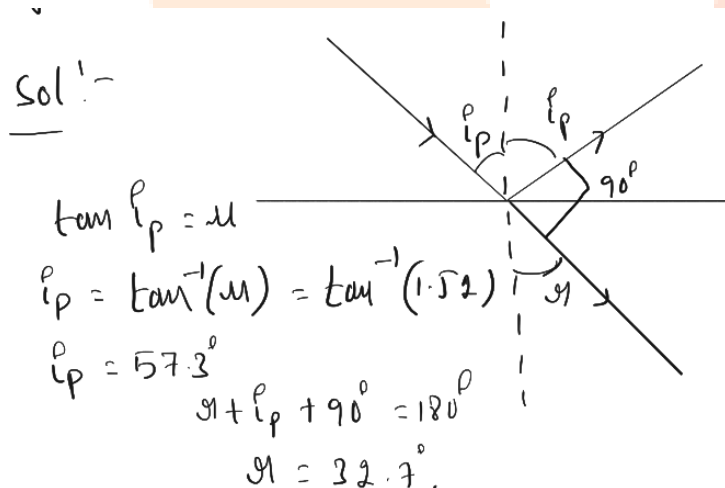
Solution:



Question: When an unpolarised light falls at a particular angle on a glass plate (placed in air). It is observed that reflected beam is completely polarised the angle of refracted beam with respect to the normal is _____
($\tan^{-1}(1.52) = 57.3^\circ$; R.I of air & glass 1 & 1.52 respectively).

Answer: (32.7°)

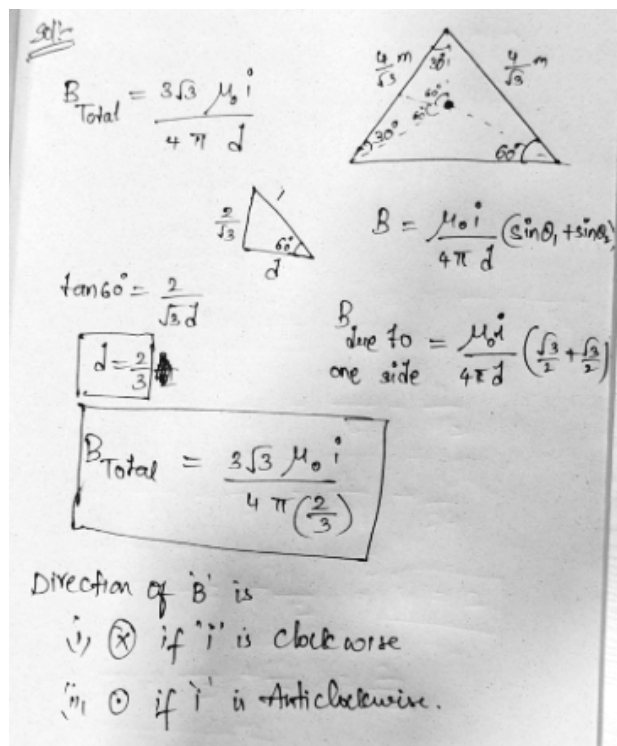
Solution:



Question: An equilateral triangle is formed by a wire of length $4\sqrt{3}\text{m}$, carrying a current of 2A. Find 'B' at centroid.

Answer: ($B_{\text{total}} = \frac{3\sqrt{3}\mu_0 i}{4\pi(\frac{2}{3})}$)

Solution:



Question: A prism has apex angle $A = 75^\circ$ and refractive index $\mu = \sqrt{3}$ (with respect to air). One of its refracting faces is coated with a thin transparent film of refractive index

$\mu_f = \sqrt{\frac{3}{2}}$. A ray is incident from air on the other (uncoated) face and enters the prism.

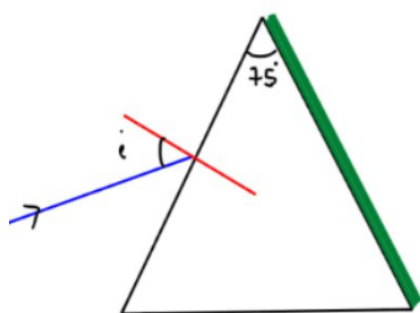
The maximum angle of incidence i_{max} on the uncoated face so that the ray undergoes total internal reflection at the coated face is:

Options:

- (a) 45°
- (b) 50°
- (c) 60°
- (d) 70°

Answer: (c)

Solution:



For TIR at prism-film interface:

$$\sin C = \frac{\mu_f}{\mu} = \frac{\sqrt{\frac{3}{2}}}{\sqrt{3}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} \Rightarrow C = 45^\circ$$

Inside the prism, if the refraction angles at the first and second faces are r_1 and r_2 :

$$r_1 + r_2 = A = 75^\circ$$

For TIR at the coated face: $r_2 \geq C = 45^\circ$

$$\Rightarrow r_1 \leq 75^\circ - 45^\circ = 30^\circ$$

At the first face (air to prism), Snell's law:

$$\sin i = \mu \sin r_1$$

Maximum i occurs at maximum allowed $r_1 = 30^\circ$:

$$\sin i_{\max} = \sqrt{3} \sin 30^\circ = \sqrt{3} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2} \Rightarrow i_{\max} = 60^\circ$$

$$i_{\max} = 60^\circ$$

Question: Four charge are kept as shown. Find the magnitude of $\vec{E}$ at point p (mid point of AB)

Options:

- (a) $\frac{45\sqrt{5}}{8}$ kV/m
(b) 180 kV/m
(c) 270 kV/m
(d) None

Answer: (a)

Solution:

Diagram showing four charges: P_1 (top left), P_2 (top right), and two charges (bottom) forming a triangle with base AB. Point P is the midpoint of AB. Distances are given as 40 cm.

Calculation of electric field E at point P:

$$E = \frac{K}{r^3} \sqrt{4P_2^2 + P_1^2}$$

$$= \frac{9 \times 10^9}{(40 \times 10^{-2})^3} \times \sqrt{4(2 \times 10^{-6})^2 + (2 \times 10^{-4})^2}$$

$$= \frac{45\sqrt{5} \times 10^3}{8} \frac{V}{m}$$

Question: A parallel plate Capacitor with plate separation 5 mm is charged by a battery on introducing a mica sheet of 2 mm and maintaining the Connections of the plates with the terminals of the battery, it is found that it draw 25% more charge from the battery The dielectric constant of mica is

Options:

- (a) 1
- (b) 2.5
- (c) 1.5
- (d) 2

Answer: (d)

Solution:

$$\begin{aligned}
 C &= \frac{\epsilon_0 A}{d} \\
 C' &= \frac{\epsilon_0 A}{d - t + t/k} \\
 &= \frac{\epsilon_0 A}{d \left(1 + \frac{t}{d} \left(\frac{1}{k} - 1\right)\right)} \\
 C' &= \frac{C}{1 + \frac{t}{d} \left(\frac{1}{k} - 1\right)} \\
 \text{As potential is same} \\
 Q &\propto C
 \end{aligned}
 \quad
 \begin{aligned}
 Q' &= \frac{5}{4} Q \\
 \frac{5Q}{4} &= \frac{5Q}{4 \left(1 + \frac{2}{5} \left(\frac{1}{k} - 1\right)\right)} \\
 4 &= 5 \left(1 + \frac{2}{5} \left(\frac{1}{k} - 1\right)\right) \\
 4 &= 5 + 2 \left(\frac{1}{k} - 1\right) \\
 -1 &= 2 \left(\frac{1}{k} - 1\right) \\
 1 - \frac{1}{k} &= \frac{1}{2} \Rightarrow \frac{1}{k} = \frac{1}{2} \Rightarrow k = 2
 \end{aligned}$$

Question: An EM waves moving with velocity 2×10^8 m/s. Given $\mu = 2\mu_0$; Find the ratio

$\frac{\epsilon}{\epsilon_0}$:

Options:

- (a) $\frac{9}{7}$
- (b) $\frac{8}{9}$
- (c) $\frac{8}{4}$
- (d) $\frac{9}{9}$

Answer: (c)

Solution:

Sol:
 We know speed of light in air (or) vacuum $c = 3 \times 10^8$,
 Given $= 2 \times 10^8$, $\mu = 2\mu_0 \rightarrow \frac{\mu}{\mu_0} = 2$
 → From Electromagnetic waves
 $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$, $v = \frac{1}{\sqrt{\mu \epsilon}}$, $\mu = \frac{\mu_0}{\epsilon_r}$
 $\epsilon_r = \frac{\epsilon}{\epsilon_0}$
 $\frac{c}{v} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \times \sqrt{\mu \epsilon}$
 $\frac{3}{2} = \sqrt{\frac{\mu}{\mu_0} \times \frac{\epsilon}{\epsilon_0}}$
 $\frac{9}{4} = 2 \times \frac{\epsilon}{\epsilon_0}$
 $\therefore \boxed{\frac{\epsilon}{\epsilon_0} = \frac{9}{8}}$

Question: At what temperature, speed of sound air will be doubled than its speed in air at 0°C ?

Options:

- (a) 1092°C
- (b) 512 K
- (c) 819 K
- (d) 1092 K

Answer: (d)

Solution:

$$T_1 = 0^\circ\text{C} = 273\text{ K}$$

Let v_2 is the doubled speed $\Rightarrow T_2 = 1092\text{ K}$ of sound at temperature T_2 . ($v_2 = 2v_1$)

As we know, speed of sound is directly proportional to the square root of temperature, we can write

$$v_1 \propto \sqrt{T_1} \dots\dots\dots (1)$$

$$\text{And } v_2 \propto \sqrt{T_2} \dots\dots\dots (2)$$

$$\text{Now, } \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

Substituting the value of v_2 in the above equation, we get,

$$\Rightarrow \frac{1}{2} = \sqrt{\frac{T_1}{T_2}}$$

$$\therefore T_1 = 273\text{ K}$$

$$\text{So, } \Rightarrow \frac{273}{T_2} = \frac{1}{4}$$

$$T_2 = 1092\text{ K}$$

① ✓

Question: One mole of an ideal gas at an initial temperature T & volume V expands to volume $2V$ via two processes

a) Isothermal

b) Isobaric

Find ratio of work done by gas in Isothermal to Isobaric.

Options:

(a) $\ln(2) : 1$

(b) $1 : \ln(2)$

(c) $R : \ln(2)$

(d) $2 : 1$

Answer: (a)

Solution:

Handwritten solution showing the calculation of work done in isothermal and isobaric processes:

$$W_{\text{isoth}} = 1RT \ln\left(\frac{2V}{V}\right) = RT \ln(2)$$

$$W_{\text{isobar}} = P(V_2 - V_1)$$

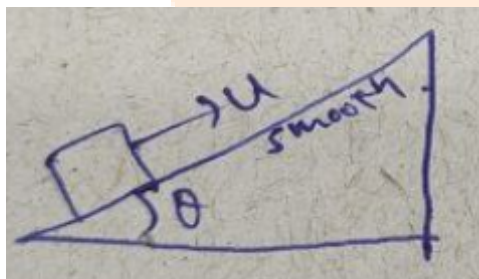
$$W_{\text{bar}} = \frac{RT}{V} (2V - V)$$

$$W_{\text{bar}} = RT$$

$$\frac{W_{\text{isoth}}}{W_{\text{isobar}}} = \frac{RT \ln(2)}{RT} = \ln(2) : 1$$

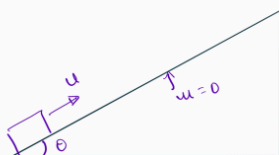
option A

Question: A body is projected up the smooth inclined plane having angle of inclination ' θ ' with the horizontal as shown in figure. Find the distance covered before stopping.



Answer: $s = \frac{u^2}{2g \sin \theta}$

Solution:



when the body is projected with 'u' velocity about smooth incline on the body given retardation acts so, that it will comes to rest after covering some displacement.

$$u^2 - 0 = 2as$$

$$u^2 = 2gs \sin \theta$$

$$s = \frac{u^2}{2g \sin \theta}$$

Question: A metallic sphere of dia 2 mm & density 10.5 g/cm^3 is dropped in glycerine having viscosity 10 poise & density 1.5 g/cm^3 . The terminal velocity attained by sphere is $x \times 10^{-2} \text{ cm/s}$; $x = ?$

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (b)

Solution:

$$V_T = \frac{2}{9} \frac{(\rho - \sigma) r^2 g}{\eta}$$

$$V_T = \frac{2}{9} \frac{(10.5 - 1.5) \left(\frac{2 \times 10^{-3}}{2} \right)^2 (10)}{10}$$

$$V_T = \frac{2}{9} (9) 10^{-2}$$

$$V_T = 2 \times 10^{-2} \text{ cm/s}$$

Question: Two point charges are fixed on the x-axis in vacuum: $q_1 = +7\mu\text{C}$ at $(-9\text{m}, 0, 0)$ and $q_2 = -2\mu\text{C}$ at $(+9\text{m}, 0, 0)$. The electrostatic potential energy of these two-charge

system is $\left(\text{take } k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \right);$

Options:

- (a) $+7 \times 10^{-3} \text{ J}$
- (b) $-7 \times 10^{-4} \text{ J}$
- (c) $-7 \times 10^{-3} \text{ J}$
- (d) $-7 \times 10^{-2} \text{ J}$

Answer: (c)

Solution:

Concise Solution

Separation between charges:

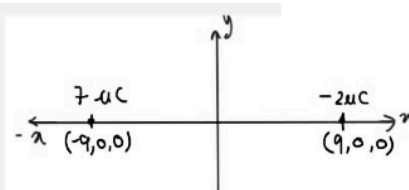
$$r = |9 - (-9)| = 18 \text{ m}$$

Potential energy of two charges:

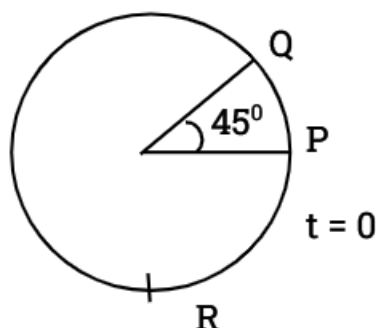
$$U = \frac{kq_1q_2}{r}$$

$$U = \frac{(9 \times 10^9)(7 \times 10^{-6})(-2 \times 10^{-6})}{18} = \frac{(9 \times 10^9)(-14 \times 10^{-12})}{18}$$

So, $U = -7 \times 10^{-3} \text{ J}.$



Question: Two particle P & Q are moving in a circular path of radius R with same angular velocity (ω). Find the ratio of time taken by P & Q to reach R.



Options:

- (a) 3 : 2
- (b) 6 : 5
- (c) 2 : 1
- (d) 4 : 3

Answer: (b)

Solution:

$$\theta = \omega t$$

$$\theta_p = 270 \quad \theta_q = 270 - 45 = 225$$

$$\frac{t_p}{t_q} = \frac{\frac{\theta_p}{\omega}}{\frac{\theta_q}{\omega}} = \frac{\theta_p}{\theta_q} = \frac{270}{225} = \frac{6}{5}$$

option b

Question: Given $i_{rms} = \frac{\alpha}{\sqrt{2}}$; $i = 10 \sin \omega t$. $\Omega = 1000$ rad/s. Find $\alpha =$

Options:

- (a) 10
- (b) $9\sqrt{2}$
- (c) $10\sqrt{2}$
- (d) $10/\sqrt{2}$

Answer: (d)

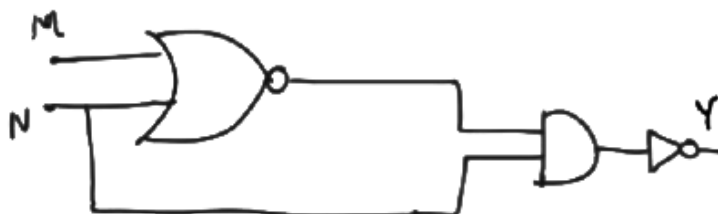
Solution:

$$i_{rms} = \frac{i_0}{\sqrt{2}} \quad \text{and} \quad i = i_0 \sin \omega t$$

$$i_0 = 10$$

$$i_{rms} = \frac{10}{\sqrt{2}} \quad \boxed{\alpha = 10}$$

Question: Truth table for given circuit will be



Options:

(a)

M	N	Y
0	0	0
0	1	0
1	0	0
1	1	0

(b)

M	N	Y
0	0	1
0	1	1
1	0	1
1	1	1

(c)

M	N	Y
0	0	0
0	1	1
1	0	1
1	1	1

(d)

M	N	Y
0	0	1
0	1	1
1	0	0
1	1	0

Answer: (b)

Solution:

$$\begin{aligned}
 \text{Sol: } Y &= \overline{(\overline{M+N}) \cdot N} \\
 Y &= \overline{\overline{M+N}} + \overline{N} \\
 Y &= (M+N) + \overline{N} \\
 &= M + (N + \overline{N}) \\
 &= M + 1 \\
 Y &= 1
 \end{aligned}$$

Question: Find dimensions of terminal velocity & if it is related as $V_t \propto g^a r^b e^c \eta^d$, then

find value of $\frac{\alpha + \beta}{\gamma - \delta} = \frac{x}{2}$. Find 'x'?

Answer: (x = 3)

Solution:

Sol We know that,

$$v_t = \frac{2}{9} \frac{(e-\sigma)r^2 g}{\eta} \rightarrow (1)$$

and Given,

$$v_t \propto g^\alpha r^\beta e^\gamma \eta^\delta \rightarrow (2)$$

From (1) & (2)

$$\alpha=1, \beta=2, \gamma=1, \delta=-1$$

Hence, $\frac{\alpha+\beta}{\gamma-\delta} = \frac{1+2}{1-(-1)} = \frac{3}{2} = \frac{n}{2}$

$$\boxed{n=3}$$

Question: Two metals with Work functions in ratio 1:2 are exposed with photons of Energy 6 eV. If $KE_A : KE_B$ is 2.642 : 1, then ϕ_A and ϕ_B values in eV are

Options:

- (a) 2.3, 4.6
- (b) 1.4, 2.8
- (c) 2.3, 3.2
- (d) 3.2, 2.3

Answer: (a)

Solution:

Q. 20) $E_p = \phi + k \cdot E_{max}$

$$E_p = 6 \text{ eV} ; \frac{\phi_A}{\phi_B} = \frac{1}{2} ; \frac{k \cdot E_A}{k \cdot E_B} = \frac{2.642}{1}$$

$$\phi_B = 2\phi_A ; k \cdot E_A = 2.642 k_B$$

$$6 = 2\phi_A + k_B$$

$$6 = \phi_A + 2.642 k_B$$

$$0 = \phi_A - 1.642 k_B$$

$$\phi_A = 1.642 k_B$$

$$6 = 2 \times 1.642 k_B + k_B$$

$$6 = 4.284 k_B$$

$$k_B = \frac{6}{4.284} = 1.4$$

$$6 = 2\phi_A + 1.4$$

$$\phi_A = \frac{4.6}{2} = 2.3 \quad \text{So } \phi_B = 4.6$$

option A.

Question: A bomb initially at rest explodes into three fragments whose masses are in the ratio 2 : 2 : 1. The two fragments of equal mass move in mutually perpendicular directions. One of these fragments has speed 18 m s⁻¹ (hence the other equal-mass fragment also has speed 18 m s⁻¹). The speed of the third fragment is:

Options:

- (a) 18√2 ms⁻¹
- (b) 36√2 ms⁻¹
- (c) 54 ms⁻¹
- (d) 72 ms⁻¹

Answer: (b)

Solution:

Solution (concise)

Let the masses be 2m, 2m, and m. Initially, total momentum is zero.

Two 2m fragments move perpendicular, each with speed 18 m s⁻¹:

$$p_1 = 2m(18) = 36m, \quad p_2 = 36m$$

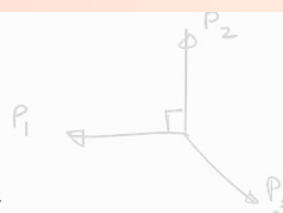
Resultant momentum of these two:

$$p_{\text{res}} = \sqrt{(36m)^2 + (36m)^2} = 36m\sqrt{2}$$

To conserve momentum, the third fragment must have equal and opposite momentum:

$$mv_3 = 36m\sqrt{2} \Rightarrow v_3 = 36\sqrt{2} \text{ m s}^{-1}$$

Correct option: (B)



Question: A circular wire of radius r lies in a plane perpendicular to magnetic field B. The loop is deformed into a square in t = 10 sec. Find the emf induced?

Answer: (0.171BL²)

Solution:

Circle

$$L = 2\pi r \quad \left| \quad A = \pi r^2 \right.$$

$$r = \frac{L}{2\pi} \quad \left| \quad A = \pi \left(\frac{L}{2\pi} \right)^2 = \frac{L^2}{4\pi} \right.$$

Square

$$L = 4s$$

$$s = \frac{L}{4}$$

$$A = s^2 = \left(\frac{L}{4} \right)^2 = \frac{L^2}{16}$$

$$\phi = B(A_f - A_i) = B \left(\frac{L^2}{16} - \frac{L^2}{4\pi} \right)$$

$$\epsilon = \frac{d\phi}{dt} = \frac{B \left(\frac{L^2}{16} - \frac{L^2}{4\pi} \right)}{10}$$

Question: A potentiometer wire of length is measured is as 100 cm. The least count of the potentiometer is '1 mm'. Find the relative error in EMF?

Options:

- (a) 1×10^{-3}
- (b) 2×10^{-3}
- (c) 3×10^{-2}
- (d) 4×10^{-1}

Answer: (a)

Solution:

$$\epsilon \propto L \Rightarrow \frac{\Delta \epsilon}{\epsilon} = \frac{\Delta L}{L}$$

$$\frac{\Delta \epsilon}{\epsilon} = \frac{0.1 \text{ cm}}{100} = 1 \times 10^{-3}$$

JEE-Main-23-01-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: In estimation of chlorine by Carius method, 0.245 g organic compound gave 0.5453 g AgCl. Find percentage of chlorine in the organic compound

Options:

- (a) 54
- (b) 32
- (c) 44
- (d) 23

Answer: (a)

Question: For XeO_2F_2 , select the correct statement(s).

- (A) It shows see-saw shape
- (B) Number of lone pair(s) of e^- on Xe is 1.
- (c) $\angle \text{FXeF} = 180^\circ$ (approx.)
- (d) It has tetrahedral shape.

Options:

- (a) (A), (C), (D) only
- (b) (A), (B) only
- (c) (A), (B), (C) only
- (d) (B), (C), (D) only

Answer: (c)

Question: Statement-I: Size of O^{2-} is smaller than F^- .

Statement-II: Electronegativity of F is more than that of oxygen.

In the light of above statements, choose the correct option

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both Statement-I and Statement-II are incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (c)

Question: Iodoform test can differentiate

Options:

- (a) Methanol & Ethanol
- (b) $\text{CH}_3 - \text{COOH}$ & $\text{CH}_3 - \text{CH}_2 - \text{COOH}$
- (c) Cyclopropene & cyclobutene
- (d) Phenol & Benzoic Acid

Answer: (a)

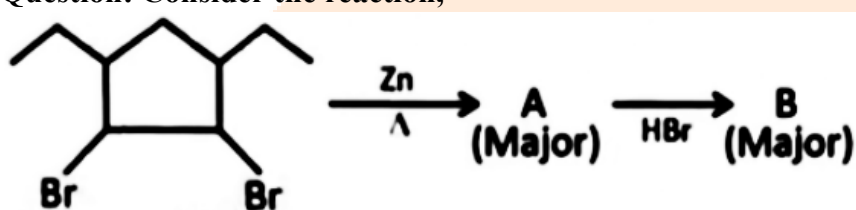
Question: The oxidation state of chromium in the final product named in the reaction between KI and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is

Options:

- (a) +2
- (b) +6
- (c) +4
- (d) +3

Answer: (d)

Question: Consider the reaction,



Choose the correct option,

Options:

- (a) A is
- (b) B is
- (c) B is
- (d) A is

(d)

Answer: (c)

Question: Both human DNA and RNA are Chiral molecules. The Chirality in DNA and RNA arise due to the presence of

Options:

- (a) Chiral Phosphate unit

- (b) D-sugar component
- (c) L-sugar component
- (d) Base unit

Answer: (b)

Question: Identify the correct set of details from the following

- A) $[\text{Co}(\text{NH}_3)_6]^{3+}$: Inner orbital complex :- d^2sp^3 hybridized
- B) $[\text{MnCl}_6]^{3-}$: outer orbital complex :- sp^3d^2 hybridized.
- C) $[\text{CoF}_6]^{3-}$: outer orbital complex :- d^2sp^3 hybridized
- D) $[\text{FeF}_6]^{3-}$: outer orbital complex :- sp^3d^2 hybridized
- E) $[\text{Ni}(\text{CN})_4]^{2-}$: Inner orbital complex :- sp^3 hybridized

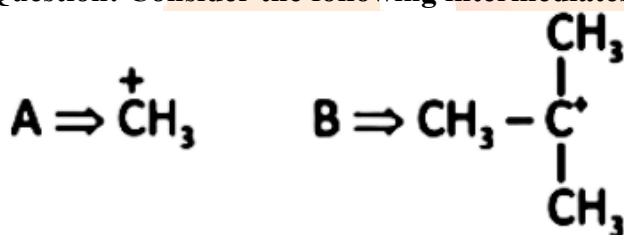
Choose the correct answer from the option given below

Options:

- (a) C, D only
- (b) A, B, & D only
- (c) A, B, C, D, E
- (d) A, C & E only

Answer: (b)

Question: Consider the following intermediates



Options:

- (a) B is more stable than A as it has 9 α hydrogen
- (b) A is more stable than B as it has 3 α hydrogen
- (c) B is more stable than A due to resonance
- (d) A is more stable due to inductive effect

Answer: (a)

Question: DNA is optically active due to the presence of

Options:

- (a) Purine nitrogen base
- (b) Phosphate molecule
- (c) D-pentose sugar
- (d) L-pentose sugar

Answer: (c)

Question: 250 cc of $x \times 10^{-3}$ M acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution neutralises 750 cc of 0.6 M Mohr's salt. Value of x is

Options:

- (a) 200
- (b) 600
- (c) 400
- (d) 300

Answer: (d)

Question: Which of the following are isobars?

Options:

- (a) ${}_{92}^{232}\text{U}$ & ${}_{92}^{238}\text{U}$
 (b) ${}_1^3\text{H}$ & ${}_1^2\text{H}$
 (c) ${}_1^3\text{H}$ & ${}_2^3\text{He}$
 (d) ${}_7^{14}\text{N}$ & ${}_7^{15}\text{N}$

Answer: (c)

Question: Two metals with work function in ratio 1 : 2, are exposed with photons of energy 6 eV. If $\text{KE}_A : \text{KE}_B$ is 2.642 : 1, then ϕ_A and ϕ_B value (in eV) are

Options:

- (a) 2.3, 4.6
 (b) 1.4, 2.8
 (c) 2.3, 3.6
 (d) 3.2, 6.4

Answer: (a)

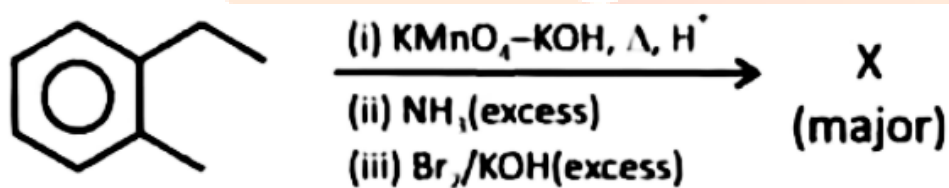
Question: An element belongs to group-15 'X', 'Y' and 'P(Phosphorus)'. The difference in electronegativity of X and P is higher than P and Y. Then X, Y are

Options:

- (a) As & Sb
 (b) N & As
 (c) Sb & N
 (d) As & N

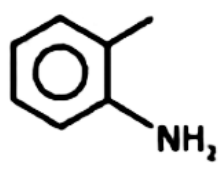
Answer: (b)

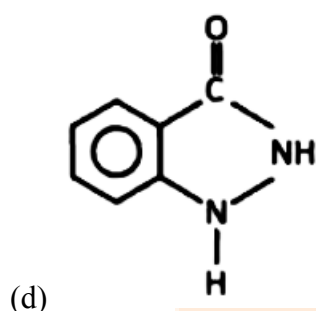
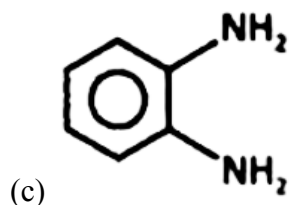
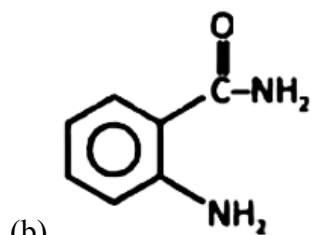
Question: Consider the reaction,



X is,

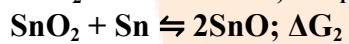
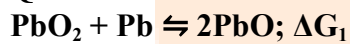
Options:

- (a) 



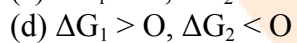
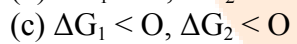
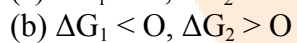
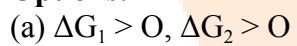
Answer: (c)

Question: Consider the following reactions



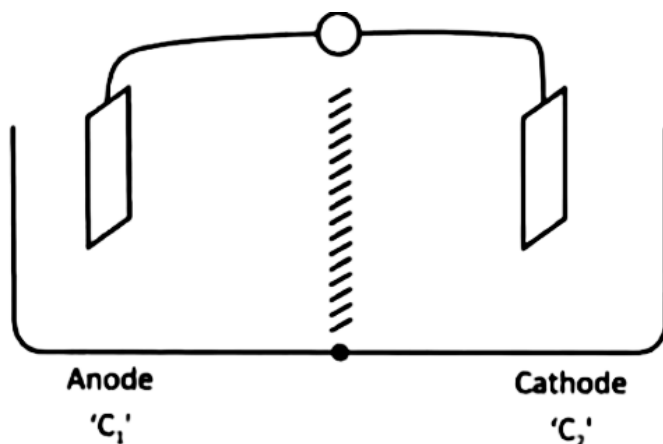
Select the correct option?

Options:



Answer: (b)

Question: Consider a galvanic cell, made up of two H_2 -electrodes,



Both compartments contain the same metal electrodes. If concentrations of H^+ in anode and cathode are C_1 and C_2 respectively, then $E_{\text{cell}} > 0$ when, ($pH_2 = 1 \text{ atm}$)

Options:

- (a) $C_2 < C_1$
- (b) $C_2 = C_1$
- (c) $C_2 > C_1$
- (d) $C_2 = 0.5 C_1$

Answer: (c)

Question: A compound P with molecular formula C_6H_7N is sparingly soluble in water. However on reaction with HCl , it becomes soluble. On reaction with $KOH + CHCl_3$, it gives foul smelling compound Q. The number of different type of H atoms present in P is

Options:

- (a) 4
- (b) 5
- (c) 7
- (d) 8

Answer: (a)

Question: Given below are two statements.

Statement-I: Potassium hypoiodite (KOI) can act as a reducing agent in reaction with H_2O_2 .

Statement-II: When $NaOCl$ reacts with KI in acidic medium, $NaOCl$ acts as oxidizing agent.

In the light of above statements, choose the correct option.

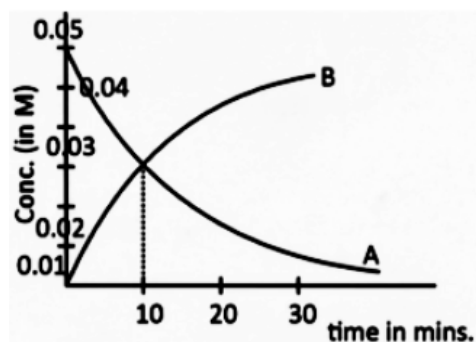
Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both Statement-I and Statement-II are incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (d)

Question: For a reaction $A_{(g)} \rightarrow nB_{(g)}$, a concentration vs time curve is,

Find 10 n.

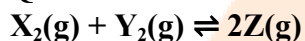


Options:

- (a) 15
- (b) 10
- (c) 18
- (d) 25

Answer: (a)

Question: Consider the following reaction



3 mol of X_2 , 3 mol of Y_2 and 9 mol of Z are present at equilibrium and the volume of container is 1L. If 10 mol of Z is added at equilibrium. Calculate the number of moles of Z at new equilibrium

Options:

- (a) 12
- (b) 15
- (c) 25
- (d) 5

Answer: (b)

Question: An ideal solution is formed by mixing 3 mole of A and 1 mole of B and the vapour pressure of solution is found to be 500 mm Hg. After further addition of 1 mole A, pressure of solution becomes 520 mm Hg. Find P°_A

Options:

Answer: (600)

JEE-Main-23-01-2026 (Memory Based)
[EVENING SHIFT]
Maths

Question: If $\sum_{k=1}^n a_k = \alpha n^2 + \beta n$ and $a_{10} = 59$, $a_6 = 79$, then find $\alpha + \beta = ?$

Answer: (5)

$$S_n = \sum a_k = \alpha n^2 + \beta n$$

$$a_n = S_n - S_{n-1}$$

$$\alpha n^2 + \beta n - (\alpha(n-1)^2 + \beta(n-1))$$

$$a_n = 2\alpha n + (\beta - \alpha)$$

$$\text{given } a_6 = 7a_1$$

$$a_{10} = 59$$

$$12\alpha + \beta - \alpha = 11\alpha + \beta$$

$$a_1 = S_1 = \alpha + \beta$$

$$a_6 = 7a_1$$

$$11\alpha + \beta = 7\alpha + 7\beta$$

$$2\alpha = 3\beta$$

$$2\alpha \cdot 10 + (\beta - \alpha) = 59$$

$$19\alpha + \beta = 59$$

Solving

$$19\alpha + \left(\frac{2}{3}\alpha\right) = 59$$

$$\frac{59\alpha}{3} = 59$$

$$\alpha = 3$$

$$2\alpha = 3\beta$$

$$\beta = 2$$

$$\text{Then } \alpha + \beta = 3 + 2 = 5$$

Question: $A = \begin{bmatrix} 0 & -2 & 3 \\ -2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$, $B(I-A) = I + A$ Then $B = ?$

Answer:

$$B = \frac{1}{3} \begin{bmatrix} -3 & 0 & 2 \\ -6 & 9 & -10 \\ -6 & 8 & -9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$B = QP^{-1}$$

$$P = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\text{adj } P = \begin{bmatrix} 0 & -3 & -3 \\ 1 & 4 & 3 \\ 1 & -5 & -3 \end{bmatrix}^T$$

$$P^T = \frac{1}{3} \begin{bmatrix} 0 & 1 & 1 \\ -3 & 4 & -5 \\ -3 & 3 & -3 \end{bmatrix}$$

$$B = \frac{1}{3} \begin{bmatrix} 1 & -2 & 3 \\ -2 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 1 \\ -3 & 4 & -5 \\ -3 & 3 & -3 \end{bmatrix}$$

$$B = \frac{1}{3} \begin{bmatrix} -3 & 0 & 2 \\ -6 & 9 & -10 \\ -6 & 8 & -9 \end{bmatrix}$$

Question: The area (in square units) between the curves $x^2 + y^2 = 4$ and $x^2 + (y-2)^2 = 4$ is

Options:

(a) $\frac{8\pi}{3} - 2\sqrt{3}$

(b) $\frac{8\pi}{3} + \sqrt{3}$

(c) $\frac{4\pi}{3} - 2\sqrt{3}$

(d) $\frac{4\pi}{3} + 2\sqrt{3}$

Answer: (a)

Area bounded by curve.

$$x^2 + y^2 = 4$$

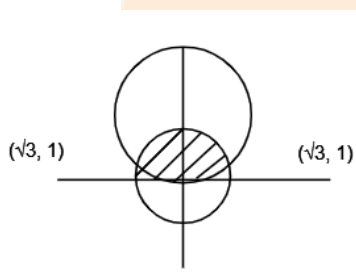
$$x^2 + (y - 2)^2 = 4$$

$$Area = 2 \int_0^{\sqrt{3}} \left[\sqrt{4 - x^2} - \left(2 - \sqrt{4 - x^2} \right) \right] dx$$

$$2 \int_0^{\sqrt{3}} \left(2 \left(\sqrt{4 - x^2} \right) - 2 \right) dx$$

$$2 \left[2 \left[\frac{x}{2} \sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} - 2x \right] \right]_0^{\sqrt{3}}$$

$$= \frac{8\pi}{3} - 2\sqrt{3}$$



Question: Number of ways to distribute 6 identical oranges among 4 persons such that each gets atleast orange is

Options:

- (a) 8
- (b) 10
- (c) 12
- (d) 13

Answer: (b)

$$n = 6$$

$$r = 4$$

$$\begin{aligned} & \binom{n-1}{r-1} \\ & \binom{6-1}{4-1} = \binom{5}{3} \\ & = \frac{5!}{3! \times 2!} \\ & = \frac{5 \times 4 \times 3!}{3! \times 2!} \\ & = 10 \end{aligned}$$

Question: The minimum value of $\cos^2 \theta + 6 \sin \theta \cos \theta + 3 \sin^2 \theta + 3$ is

Options:

- (a) -1
- (b) 1
- (c) $5 + \sqrt{10}$
- (d) $5 - \sqrt{10}$

Answer: (d)

$$\begin{aligned} & \cos^2 \theta + 6 \sin \theta \cos \theta + 3 \sin^2 \theta + 3 \\ & \frac{1+\cos 2\theta}{2} + 3 \sin 2\theta + 3 \left(\frac{1-\cos 2\theta}{2} \right) + 3 \\ & \left(\frac{1}{2} + \frac{3}{2} + 3 \right) + \left(\frac{1}{2} \cos 2\theta - \frac{3}{2} \cos 2\theta \right) + 3 \sin 2\theta \\ & 5 - \cos 2\theta + 3 \sin 2\theta \\ & -\sqrt{3^2 + (-1)^2} \leq -\cos 2\theta + 3 \sin 2\theta \leq \sqrt{3^2 + (-1)^2} \\ & (+5) \\ & 5 - \sqrt{10} \leq 5 - \cos 2\theta + 3 \sin 2\theta \leq 5 + \sqrt{10} \\ & \text{min value} = 5 - \sqrt{10} \end{aligned}$$

Question: Let $A = \{1, 2, 3, \dots, 9\}$; xRy if $x - y$ is multiple of 3.

S_1 : Number of elements in R is 36

S_2 : R is equivalence relation

Options:

- (a) S_1 & S_2 both are correct
- (b) S_1 is correct but S_2 is not correct
- (c) S_2 is correct but S_1 is not correct
- (d) S_1 & S_2 both are incorrect.

Answer: (c)

$$(3k_1, 3k_2), (3k_1 + 1, 3k_2 + 1), (3k_1 + 2, 3k_2 + 2)$$

$$\text{If } x \in (3k_1, 3k_2)$$

$$\text{No. of choices } 3 \times 3 + 3 \times 3 + 3 \times 3 = 27$$

$$\text{Reflexive } \forall (a, a) \in R \Rightarrow R \text{ is Reflexive}$$

$$\text{Symmetric } (a, b) \in R \text{ and } (b, a) \in R$$

$$\text{Multiplies } 3$$

$$\text{Transitive Relation}$$

$$(x - y) \text{ is a multiple of } 3 \text{ then}$$

$$(y - z) \text{ is also multiple of } 3$$

$$\Rightarrow (x - z) \text{ is also multiple of } 3$$

$$S_1 \text{ is incorrect}$$

$$S_2 \text{ is correct.}$$

Question: Bag A contains 9w & 8B and Bag B contains 6w & 4B balls A ball is randomly transferred from bag B to Bag A then a ball is drawn from A If $P(\text{drawn balls is white from bag A}) = p/q$, p, q , are coprime, find $p+q$.

Options:

$$(a) 21$$

$$(b) 23$$

$$(c)$$

$$(d)$$

Answer: (b)

$$\text{Bag A : - Bag B : -}$$

$$9 \text{ white} \quad 6 \text{ white}$$

$$8 \text{ black} \quad 4 \text{ black}$$

$$\text{Prob of transvers white ball } P(T_w) = \frac{6}{10} = \frac{3}{5}$$

$$,, \quad black = P(T_b) = \frac{4}{10} = \frac{2}{5}$$

$$\text{If } T_w \text{ Now Bag A } \Rightarrow 9 + 1 = 10 \text{ white}$$

$$+ 8 \text{ black}$$

$$= 18 \text{ in total}$$

$$\text{If } T_b \text{ Now Bag B } \Rightarrow 9 \text{ white}$$

$$8 + 1 = 9 \text{ black}$$

$$= 18 \text{ in total}$$

$$P(\text{drawing white from after transfers})$$

$$P(D_w) = \frac{10}{18} \times \frac{3}{5} + \frac{9}{18} \times \frac{2}{5} = \frac{1}{3} + \frac{1}{5} = \frac{5+3}{15} = \frac{8}{15} = \frac{p}{q}$$

$$p + q = 8 + 15 = 23$$

Question: The sum of all the real solutions of equation

$$\log_{(x+3)}(6x^2+28x+30) = 5 - 2\log_{(6x+10)}(x^2+6x+9) \text{ is equal to}$$

Options:

- (a) 2
(b) 4
(c) 1
(d) 0

Answer: (d)

$$\log_{x+3}^{6x^2+128x+30} = 5 - 2 \log_{6x+10}^{x^2+6x+9}$$

$$\log_{x+3}^{2(3x^2+14x+15)} = 5 - 2 \log_{2(3x+5)}^{(x+3)^2}$$

$$\log_{x+3}^{2(3x+5)(x+3)} = 5 - 4 \log_{2(3x+5)}^{x+3}$$

$$\log_{x+3}^2 + \log_{x+3}^{3x+5} + \log_{x+3}^{x+3} = 5 - 4 \frac{\log x+3}{\log 2+\log 3x+5}$$

$$\frac{\log^2}{\log_{x+3}} + \frac{\log^{3x+5}}{\log_{x+3}} + 4 \frac{1}{\left(\frac{\log^2}{\log_{x+3}} + \frac{\log^{3x+5}}{\log_{x+3}}\right)} - 4 = 0$$

$$\text{Let } y = \log_{x+3}^{3x+5}; z = \log_{x+3}^2$$

$$z + y + 4 \left(\frac{1}{z+y} \right) - 4 = 0$$

$$\text{let } u = z + y$$

$$u + \frac{4}{u} - u = 0 \Rightarrow u^2 - 4u + 4 > 0$$

$$(u - 2)^2 = 0$$

$$u = 2$$

$$z + y = 2 \Rightarrow \frac{\log^2 + \log^{3x+5}}{\log x+3} = 2$$

$$\frac{\log^{2(3x+5)}}{\log x+3} = 2 \Rightarrow \log_{x+3}^{6x+10} = 2$$

$$6x + 10(x + 3)^2 = x^2 + 9 + 64$$

$$x^2 = 1 \Rightarrow x = +1 \& -1$$

$$\text{Sum} = +1 - 1 = 0$$

Question: S= No. of 4- digit numbers abcd where product of digits is 20

P= No. of 5-digit number abcde where product of digits is 20, then S + P is equal to

Answer: (74)

Possible Number of 4 digits

$$\left\{ 5, 4, 1, 1 \Rightarrow \frac{4!}{2!} = 12 \right.$$

$$\left\{ 5, 2, 2, 1 \Rightarrow \frac{4!}{2!} = 12 \right.$$

$$P \rightarrow \{5, 4, 1, 1\} \Rightarrow \frac{5!}{3!} = 20$$

$$\{1, 1, 2, 2, 5\} \Rightarrow \frac{5!}{2! \times 2!} = 30$$

$$S + P$$

$$20 + 30 + 12 + 12 = 74$$

Question: If the points of intersection of the ellipses

$x^2 + 2y^2 - 6x - 12y + 23 = 0$ and $4x^2 + 2y^2 - 20x - 12y + 35 = 0$ lie on a circle of radius r and centre (a, b) then find the value of $ab + 18r^2$

Options:

- (a) 51
- (b) 52
- (c) 55
- (d) 53

Answer: (c)

$$18r^2 = 18\left(\frac{64}{9} + 9 - \frac{27}{2}\right)$$

$$= 2(64) + 162 - 9(27)$$

$$= 290 - 243$$

$$18r^2 = 47$$

$$ab + 18r^2 = 8 + 47 = 55$$

$$ab + 18r^2 = 55$$

Question: Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c})$. If $|\vec{a}| = 1$, $|\vec{b}| = 4$, $|\vec{c}| = 2$, and the angle between $\vec{b}$ and $\vec{c}$

is 60° , then $|\vec{a} \cdot \vec{c}|$ is equal to

Answer: (1)

$$\vec{a} \times \vec{b} - 2(\vec{a} \times \vec{c}) = 0$$

$$\vec{a} \times (\vec{b} - 2\vec{c}) = 0$$

$$\vec{b} - 2\vec{c} = \lambda \vec{a}$$

$$\vec{b} = 2\vec{c} + \lambda \vec{a}$$

$$|b - 2c|^2 = |\lambda \vec{a}|^2$$

$$16 + 16 - 4 \times 4 \times 2 \times \frac{1}{2} = \lambda^2 \times 1$$

$$\lambda^2 = 16$$

$$\lambda = \pm 4$$

$$\bar{b} \cdot \bar{c} = 2c^2 \pm 4\bar{a} \cdot \bar{c}$$

$$4 = 8 \pm 4\bar{a} \cdot \bar{c}$$

$$\bar{a} \cdot \bar{c} = \pm 1$$

$$|\bar{a} \cdot \bar{c}| = 1$$

Question: If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}$ then $(z^{201} - i)^8$ is

Options:

- (a) 0
- (b) 256
- (c) 1
- (d) -1

Answer: (b)

$$z = \frac{\sqrt{3}}{2} + \frac{i}{2}$$

$$z = \cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right)$$

$$z = e^{i\frac{\pi}{2}}$$

$$(z^{201} - i)^8 = ?$$

$$z^{201} = \left(e^{i\frac{\pi}{6}}\right)^{201} = e^{i\frac{201}{6}\pi}$$

$$z^{201} = e^{i(33\pi + \frac{\pi}{2})}$$

$\therefore 33\pi$ odd multiple of π

$$z^{201} = e^{i(\frac{3\pi}{2})}$$

$$z^{201} = \cos\left(\frac{3\pi}{2}\right) + i \sin\left(\frac{3\pi}{2}\right)$$

$$= 0 - i$$

$$z^{201} = -i$$

$$\text{Find } (2^{201} - i)^8$$

$$\Rightarrow (-i - i)^8$$

$$\Rightarrow (-2i)^8$$

$$\Rightarrow (-2)^8 \cdot i^8$$

$$= 256 \times 1$$

$$= 256$$

$$\cot \theta = \frac{-1}{2\sqrt{2}} \text{ where } \theta \in \left(\frac{3\pi}{2}, 2\pi\right) \text{ then value of}$$

Question: If $\sin \frac{15\theta}{2} (\sin 8\theta + \cos 18\theta) + \cos \frac{15\theta}{2} (\cos 8\theta - \sin 8\theta)$ is _____

Options:

- (a) $\frac{\sqrt{2} + 1}{\sqrt{3}}$
 (b) $\frac{-(\sqrt{2} + 1)}{\sqrt{3}}$
 (c) $\frac{-(\sqrt{2} - 1)}{\sqrt{3}}$
 (d) $\left(\frac{\sqrt{2} - 1}{\sqrt{3}}\right)$

Answer: (a)

$$\cot \theta = -\frac{1}{2\sqrt{2}} \quad \theta \in \left(\frac{3\pi}{2}, 3\pi\right) \in \theta$$

$$\sin \frac{15\theta}{2} (\sin 8\theta + \cos 8\theta) + \cos \frac{15\theta}{2} (\cos 8\theta - \sin 8\theta) = 1$$

$$\tan \theta = -2\sqrt{2}; \tan^2 \theta = 8$$

$$\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + 8} = \sqrt{9} = 3$$

$$\cos \theta = \frac{1}{3} \quad (\theta \in \theta_4) \text{ so } +ve$$

$$\sin \theta = -\sqrt{1 - \cos^2 \theta} = -\sqrt{1 - \frac{1}{9}} = -\sqrt{\frac{8}{9}}$$

$$\sin \theta = -\frac{2\sqrt{2}}{3} \quad (\sin \text{ is } -ve \text{ in } \theta_4)$$

$$\sin \frac{15\theta}{2} \sin 8\theta + \sin \frac{15\theta}{2} \cdot \cos 8\theta + \cos 8\theta - \cos \frac{15\theta}{2} \sin 8\theta$$

rearrange

$$\left(\sin \frac{15\theta}{2} \cos 8\theta - \cos \frac{15\theta}{2} \cdot \sin 8\theta \right) + \left(\cos \frac{15\theta}{2} \cos 8\theta \sin \frac{15\theta}{2} \cdot \cos 8\theta \right)$$

$$= \sin \left(\frac{15\theta}{2} - 8\theta \right) + \cos \left(\frac{15\theta}{2} - 8\theta \right)$$

$$= \sin \left(\frac{-\theta}{2} \right) + \cos \left(\frac{-\theta}{2} \right)$$

$$= -\sin \frac{\theta}{2} + \cos \frac{\theta}{2}$$

$$\cos \theta - \frac{1}{3} \Rightarrow \cos \frac{\theta}{2} = -\sqrt{\frac{1+\frac{1}{3}}{2}} = -\sqrt{\frac{4}{6}} = -\frac{\sqrt{2}}{\sqrt{3}}$$

$$\sin \frac{\theta}{2} = +\sqrt{\frac{1-\frac{1}{3}}{2}} = \sqrt{\frac{2}{6}} = \frac{1}{\sqrt{3}}$$

$$-\frac{1}{\sqrt{3}} - \frac{\sqrt{2}}{\sqrt{3}} = -\left(\frac{\sqrt{2}+1}{\sqrt{3}} \right)$$

Question: The value of $\int \frac{dx}{(4x+6)\sqrt{4x^2+12x+7}}$ is equal to
Options:

(a) $\frac{\sqrt{2}}{8} \tan^{-1} \left(\frac{4x^2+12x+7}{\sqrt{2}} \right) + c$

(b) $\frac{\sqrt{2}}{8} \tan^{-1} (4x^2+12x+7) + c$

(c) $\frac{1}{2} \tan^{-1} \left(\frac{4x^2+12x+7}{2} \right) + c$

(d) $\frac{\sqrt{2}}{4} \tan^{-1} \left(\frac{4x^2+12x+7}{\sqrt{2}} \right) + c$

Answer: (d)

$$I = \int \frac{dx}{(4x+6)\sqrt{4x^2+12x+7}}$$

$$I = \int \frac{dx}{2(2x+3)\sqrt{(2x+3)^2-9}}$$

Let $u = 2x + 3$

$$du = 2dx \Rightarrow dx = \frac{du}{2}$$

$$I = \frac{1}{4} \int \frac{du}{u\sqrt{u^2 - (\sqrt{2})^2}}$$

$$I = \frac{1}{4} \left[\frac{1}{\sqrt{2}} \sec^{-1} \left(\frac{u}{\sqrt{2}} \right) \right] + c$$

$$I = \frac{1}{4\sqrt{2}} \sec^{-1} \left(\frac{u}{\sqrt{2}} \right) + c \text{ let } y = \frac{u}{\sqrt{2}}$$

$$\therefore \text{Identify } \sec^{-1} y = \tan^{-1}(\sqrt{y^2 - 1})$$

$$= \tan^{-1} \sqrt{\frac{4^2 - 2}{2}}$$

$$\therefore u = 2x + 3 = \tan^{-1}\left(\sqrt{\frac{4^2 - 2}{\sqrt{2}}}\right)$$

$$= \tan^{-1}\left(\sqrt{\frac{(2x+3)^2 - 2}{\sqrt{2}}}\right)$$

$$= \tan^{-1}\left(\sqrt{\frac{4x^2 + 12x + 7}{\sqrt{2}}}\right)$$

$$I = \frac{1}{4\sqrt{2}} \tan^{-1}\left(\sqrt{\frac{4x^2 + 12x + 7}{\sqrt{2}}}\right)$$

$$I = \frac{\sqrt{2}}{8} \tan^{-1}\left(\sqrt{\frac{4x^2 + 12x + 7}{\sqrt{2}}}\right) + c$$

Question: $y = f(x)$ satisfies the differential equation $(x^2 - 4)y' - 2xy + 2x(4 - x^2)^2 = 0$ and $f(3) = 15$, then find local maximum value of $f(x)$

Options:

- (a) 16
- (b) 20
- (c) 25
- (d) 30

Answer: (a)

$$(x^2 - 4)y' - 2xy + 2x(4 - x^2)^2 = 0$$

$$(x^2 - 4) \frac{dy}{dx} - 2xy = -2x(4 - x^2)^2$$

$$\frac{dy}{dx} - \frac{2x}{x^2 - 4}y = -2x(4 - x^2)$$

$$\frac{dy}{dx} - \frac{2x}{x^2 - 4}y = -2x^3 + 8x$$

$$IF = e^{\int -\frac{2x}{x^2 - 4} dx}$$

$$\text{let } u = x^2 - 4$$

$$du = 2x dx$$

$$IF = e^{-\int \frac{1}{u} du} = e^{-\ln|u|} = \frac{1}{|x^2 - 4|}$$

$$IF = \frac{1}{x^2 - 4}$$

$$y \cdot \frac{1}{(x^2 - 4)} = \int \left(-2x(x^2 - 4) \times \frac{1}{x^2 - 4} \right) dx$$

$$\frac{y}{x^2 - 4} = \int -2x dx$$

$$\frac{y}{x^2 - 4} = -x^2 + c$$

$$y = (x^2 - 4)(-x^2 + c)$$

$$f(3) = 15$$

$$15(3^2 - 4)(-3^2 + c)$$

$$15 = 5(-9 + c)$$

$$c = 12$$

$$f(x) = (x^2 - 4)(12 - x^2)$$

Let $t = x^2$

$y = (t - 4)(12 - t)$

$y = -t^2 + 16t - 48$

Local minimum

$F(x) = -x^4 + 16x^2 - 48$

$f'(x) = -4x^3 + 32x$

$f'(x) = 0$

$-4x(x^2 - 8) = 0$

$x = 0, x = \pm\sqrt{8}$

At $x = 0$; $f(0) = -48$ (local min)

At $x = \pm\sqrt{8}$; $x^2 = 8$

$f(\pm\sqrt{8}) = -(8)^2 + 16(8) - 48$

$= -64 + 128 - 48$

$f(\pm\sqrt{8}) = 16$

Question: An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex 'O' at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is.

Options:

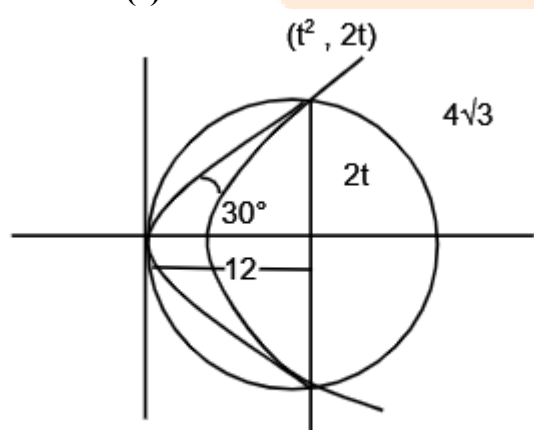
(a) $2(3 + \sqrt{3})$

(b) $4(6 + \sqrt{3})$

(c) $4(3 - \sqrt{3})$

(d) $2(8 - 3\sqrt{3})$

Answer: (c)



$$= \sqrt{11} \frac{(\sqrt{5}+9)}{-80}$$

$$\tan 30 = \frac{2^+}{t^2}$$

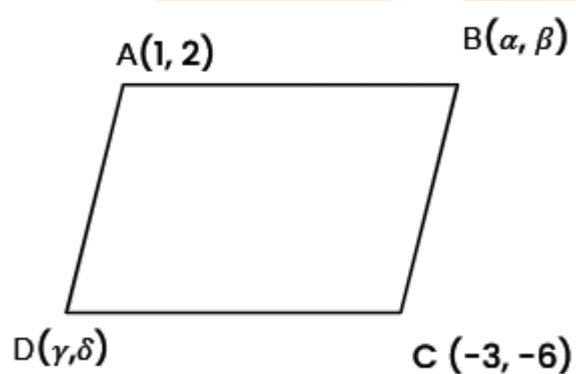
$$\frac{1}{\sqrt{3}} = \frac{2}{t}$$

$$t = 2\sqrt{3}$$

$$12 - 4\sqrt{3}$$

$$= 4(3 - \sqrt{3})$$

Question: Let A(1, 2) and C (-3, -6) be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x - y = 14$. If B(α, β) and D(γ, δ) are the other two vertices, then $|\alpha + \beta + \gamma + \delta|$ is equal to.



Options:

- (a) 6
- (b) 1
- (c) 9
- (d) 3

Answer: (a)

$A(1, 3) B(\alpha, \beta), C(-3, -6) D(\gamma, \delta)$

mid point of AC = midpoint of BD

$$m = \left(\frac{1-3}{2}, \frac{2-6}{2} \right) = (-1, -2)$$

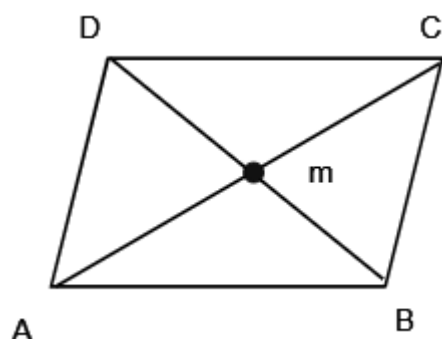
$$m = \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2} \right)$$

$$\frac{\alpha+\gamma}{2} = -1 \Rightarrow \alpha + \gamma = -2$$

$$\frac{\beta+\delta}{2} = -2 \Rightarrow \beta + \delta = -4$$

$$(\alpha + \gamma) + (\beta + \delta) = -2 - 4 = -6$$

$$|\alpha + \beta + \gamma + \delta| = |-6| = 6$$



Question: If the mean and the variance of the data

Class	4 - 8	8 - 12	12 - 6	16 - 20
Frequency	3	λ	4	-7

are μ and 19 respectively, then the value of $\lambda + m$ is

Options:

- (a) 18
- (b) 21
- (c) 19
- (d) 20

Answer: (c)

$$N = 20$$

$$\text{Sum of frequency} = N = 3 + \lambda + 4 + 7$$

$$20 = 14 + \lambda$$

$$\lambda = 6$$

$$\text{Mean}(m) = A + h \left(\sum \frac{F_i u_i}{M} \right)$$

$$m = 10 + 4 \left(\frac{15}{20} \right)$$

$$m = 10 + 4(0.15)$$

$$m = 13$$

$$\lambda + m = 6 + 13 = 19$$

$$\lambda + m = 19$$

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

